

PHYSICS

PAPER 1 (THEORY)

Maximum Marks: 70**Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 9 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given after these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1.	Mass of an electron	m_e	=	$9.1 \times 10^{-31} \text{ kg}$
2.	Charge of an electron (magnitude)	e	=	$1.6 \times 10^{-19} \text{ C}$
3.	Planck's constant	h	=	$6.6 \times 10^{-34} \text{ J s}$
4.	Speed of light in vacuum	c	=	$3 \times 10^8 \text{ m s}^{-1}$
5.	Permittivity of vacuum	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
6.		$1/4\pi\epsilon_0$	=	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
7.	One electron volt	1 eV	=	$1.6 \times 10^{-19} \text{ J}$
8.	Atomic mass unit	1 u	=	$931.5 \text{ MeV}/c^2 = 1.66 \times 10^{-27} \text{ kg}$
9.	Rydberg constant	R	=	$1.097 \times 10^7 \text{ m}^{-1}$
10.	Acceleration due to gravity	g	=	9.8 m s^{-2}
11.	Mass of a hydrogen atom	m_H	=	1.007825 u
12.	Mass of a neutron	m_n	=	1.008665 u

Instruction to Supervising Examiner: Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS

Question 1

- (A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.
- (i) A hollow metal sphere of radius R carries a charge Q . At a point inside it, at a distance $R/2$ from the centre: (*Understand*) [1]
- (a) $E = 0$ and $V = 0$
 - (b) $E = 0$ and $V = kQ/R$
 - (c) $E = kQ/R^2$ and $V = kQ/R$
 - (d) $E = 0$ and $V = 2kQ/R$
- (ii) Identify the correct statements about magnetic materials. [1]
- (P) The magnetic susceptibility of a diamagnetic substance is negative.
 - (Q) The susceptibility of a paramagnetic substance is inversely proportional to its absolute temperature.
 - (R) Above its Curie temperature a ferromagnetic substance becomes diamagnetic.
 - (S) The relative permeability of a diamagnetic substance is greater than 1.
- (*Understand*)
- (a) Both (P) and (Q)
 - (b) Both (P) and (S)
 - (c) Both (Q) and (R)
 - (d) Both (R) and (S)
- (iii) A proton passes undeflected through a region in which a uniform electric field of $2 \times 10^4 \text{ V m}^{-1}$ and a uniform magnetic field of 0.1 T are perpendicular to each other and to its velocity. The speed of the proton is: (*Apply*) [1]
- (a) $2 \times 10^3 \text{ m s}^{-1}$
 - (b) $2 \times 10^5 \text{ m s}^{-1}$
 - (c) $5 \times 10^{-6} \text{ m s}^{-1}$
 - (d) $2 \times 10^4 \text{ m s}^{-1}$
- (iv) Select the odd one out: the observation that does **not** support the photon (particle) picture of light. (*Analyse*) [1]
- (a) Existence of a threshold frequency for photoemission
 - (b) Emission of photoelectrons without any time lag
 - (c) Maximum kinetic energy of photoelectrons independent of intensity
 - (d) Interference fringes in Young's double-slit experiment

- (v) Match the entries in Column I with those in Column II. [1]

Column I	Column II
1. LED	A. Voltage regulator
2. Photodiode	B. Converts electrical energy into light
3. Solar cell	C. Detects light when reverse biased
4. Zener diode	D. Converts light energy into electrical energy with no bias

(Recall)

- (a) 1-B, 2-C, 3-D, 4-A
 (b) 1-C, 2-B, 3-A, 4-D
 (c) 1-B, 2-D, 3-C, 4-A
 (d) 1-A, 2-C, 3-D, 4-B
- (vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: A convex lens of glass ($n = 1.5$) behaves as a diverging lens when immersed in carbon disulphide ($n = 1.63$).

Reason: The focal length of a lens depends on the refractive index of its material relative to the surrounding medium.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.
- (vii) In Young's double-slit experiment, the fifth dark fringe from the centre is formed at a point where the path difference between the two waves is: (Apply) [1]
- (a) 5λ
 (b) $5\lambda/2$
 (c) $9\lambda/2$
 (d) $11\lambda/2$

(B) Answer the following questions briefly:

- (i) Why can two equipotential surfaces never intersect? (Understand) [1]
 (ii) What is the power factor of a circuit containing only a pure inductor? What is the average power consumed? (Recall) [1]
 (iii) Name the phenomenon on which an optical fibre works, and state the two conditions for it. (Recall) [1]
 (iv) The de Broglie wavelength of an electron is λ . What will it become if the kinetic energy of the electron is made four times as large? (Apply) [1]
 (v) Why must the neutrons produced in the fission of uranium-235 be slowed down in a reactor? (Understand) [1]

- (vi) What happens to the width of the depletion layer of a p-n junction when it is forward biased? [1]
(Recall)
- (vii) Fill in the blank with an appropriate term from the bracket. [1]
A ray of light passing through the _____ (principal focus, optical centre, centre of curvature, pole) of a thin lens emerges undeviated. (Recall)

SECTION B – 14 MARKS

Question 2

- (i) Each key of a computer keyboard acts as a parallel-plate capacitor. The plates of one key have an area of 2 cm^2 and are 1 mm apart, with a material of dielectric constant 5 between them. Calculate its capacitance and the charge on it when a potential difference of 3 V is applied. [2]
(Apply)

OR

- (ii) An oil drop of mass $3.2 \times 10^{-15} \text{ kg}$ is held stationary between two horizontal plates in a uniform electric field of $9.8 \times 10^4 \text{ N C}^{-1}$ directed vertically downwards. Find the charge on the drop and the number of excess electrons on it. (Apply) [2]

Question 3

- Draw a labelled circuit diagram of a metre bridge used to find an unknown resistance. Write the formula used. (Understand) [2]

Question 4

- (i) Figure 1 shows the I - V graphs of a metallic wire at two temperatures T_1 and T_2 . Which temperature is higher? Justify your answer. (Analyse) [2]

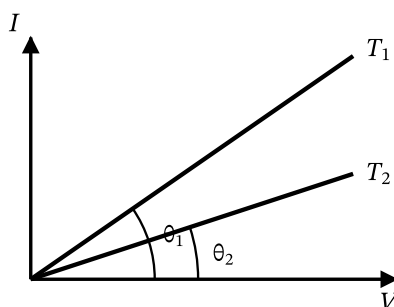


Figure 1

OR

- (ii) A wire of resistance 20Ω is bent into a circle. Find the resistance between two diametrically opposite points. What would the resistance of the original wire be if it were instead stretched uniformly to twice its length? (Apply) [2]

Question 5

- A square loop of side 10 cm and resistance 0.5Ω is placed with its plane perpendicular to a magnetic field. The field decreases steadily from 0.2 T to zero in 0.1 s. Calculate the induced emf and the induced current. (Apply) [2]

Question 6

- (i) An electromagnetic wave has a frequency of 10^{11} Hz. Name the wave and give one use of it. [1]
(Apply)
- (ii) Write an expression for the speed of electromagnetic waves in vacuum in terms of μ_0 and ϵ_0 . [1]
State one property common to X-rays, red light and radio waves. (Recall)

Question 7

- (i) Distinguish between an intrinsic and an extrinsic semiconductor. (Understand) [1]
- (ii) An n-type semiconductor has a large number of free electrons. Is it negatively charged? [1]
Explain. (Understand)

Question 8

Green light ejects photoelectrons from a metal surface but yellow light does not. Will (i) blue [2]
light and (ii) red light eject photoelectrons from it? Justify your answers. (Analyse)

SECTION C – 27 MARKS**Question 9**

- (i) Using Gauss's theorem, find the electric field intensity due to a uniformly charged thin [3]
spherical shell of radius R at a point (a) outside it and (b) inside it. Draw a graph of E against
distance r from the centre. (Understand)

OR

- (ii) Three capacitors of capacitances C_1 , C_2 and C_3 are connected in series. Derive an expression [3]
for the equivalent capacitance of the combination. (Understand)

Question 10

- (i) In the circuit of Figure 2, a 12 V cell of internal resistance $1\ \Omega$ in series with a $3\ \Omega$ resistor, and a 6 V cell of internal resistance $0.5\ \Omega$ in series with a $1.5\ \Omega$ resistor, are connected in parallel across a $2\ \Omega$ resistor. Using Kirchhoff's laws, find the current in each branch. (*Apply*) [3]

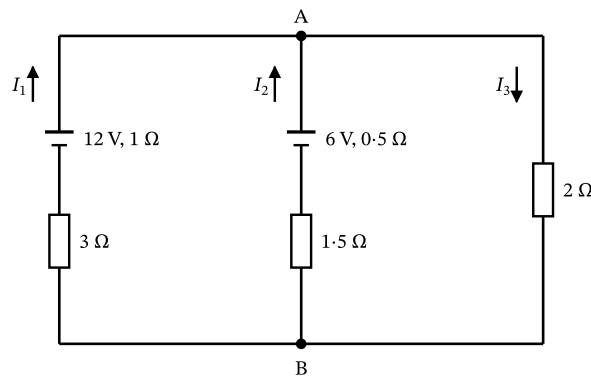


Figure 2

OR

- (ii) Figure 3 shows the terminal voltage V of a cell plotted against the current I drawn from it. From the graph find (a) the emf of the cell and (b) its internal resistance. (c) What current will it send through an external resistance of $2.5\ \Omega$? (*Analyse*) [3]

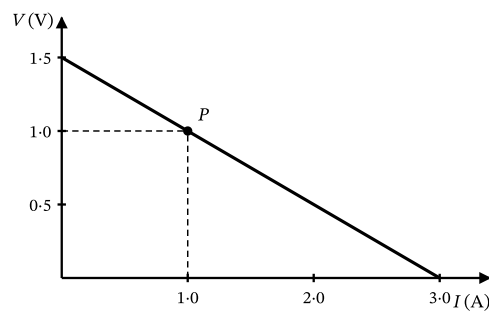


Figure 3

Question 11

- Derive an expression for the force per unit length between two long, straight, parallel conductors carrying currents I_1 and I_2 , a distance r apart. Hence define one ampere. (*Understand*) [3]

Question 12

- (i) A galvanometer of resistance $60\ \Omega$ gives full-scale deflection for 5 mA. Calculate the shunt needed to convert it into an ammeter of range 0–3 A, and the resistance of the ammeter. (*Apply*) [2]
- (ii) Why is the magnetic field in a moving-coil galvanometer made radial? (*Recall*) [1]

Question 13

During a practical class, a teacher asked the following questions. State whether each student's answer is correct or incorrect. Correct an incorrect response with a brief explanation.

- (i) **Teacher:** With a convex lens, a sharp image is formed at 30 cm when the object is at 30 cm. [1]
What is the focal length?
Student: 30 cm. (*Evaluate*)
- (ii) **Teacher:** The lens forms a sharp image of a distant tree on a screen 20 cm away. What is its [1]
power?
Student: +5 D. (*Evaluate*)
- (iii) **Teacher:** Is the angle of minimum deviation of a glass prism the same for red and violet light? [1]
Student: Yes, it depends only on the angle of the prism. (*Evaluate*)

Question 14

Using Huygens' principle, draw a labelled diagram of a plane wavefront reflected at a plane [3]
surface and prove that the angle of incidence equals the angle of reflection. (*Understand*)

Question 15

Light of wavelength 600 nm falls normally on a single slit of width 0.2 mm. The diffraction [3]
pattern is observed on a screen 1.5 m away.

- (i) Draw a graph showing how the intensity varies with angle in the pattern. (*Understand*) [1]
(ii) Calculate the angular width and the linear width of the central maximum. (*Apply*) [1]
(iii) How does the width of the central maximum change if the slit is made narrower? Why? [1]
(*Analyse*)

Question 16

Figure 4 shows some energy levels of the hydrogen atom. A photon emitted in the transition [3]
from $n = 3$ to $n = 1$ falls on a metal surface of work function 4.2 eV.

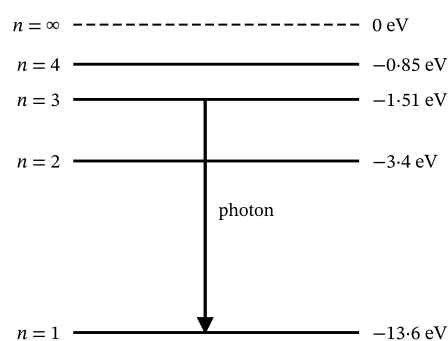


Figure 4

- (i) What is the energy of the photon? (*Apply*) [1]
(ii) Calculate its wavelength. (*Apply*) [1]
(iii) Find the stopping potential for the photoelectrons emitted. (*Apply*) [1]

Question 17

- (i) Draw energy band diagrams to distinguish between a conductor, a semiconductor and an [1]
insulator. (*Recall*)

- (ii) Figure 5 shows the I - V characteristic of a junction diode. Find its dynamic resistance between 0.7 V and 0.8 V. (*Apply*) [1]

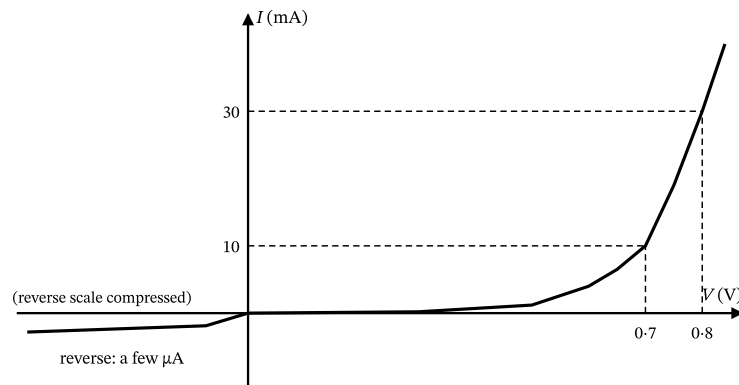


Figure 5

- (iii) Why does the reverse current of the diode remain almost constant as the reverse voltage is increased (below breakdown)? (*Understand*) [1]

SECTION D – 15 MARKS

Question 18

- (i) (a) A series circuit contains a resistance of $20\ \Omega$, an inductor of $0.1\ \text{H}$ and a capacitor of $10\ \mu\text{F}$, connected to a $200\ \text{V}$ (rms) source of variable frequency. Calculate (1) the resonant frequency, (2) the rms current at resonance and (3) the rms voltage across the inductor at resonance. (*Apply*) [3]
- (b) Sketch graphs of rms current against frequency for this circuit for two values of the resistance, $R_1 < R_2$. Which gives the sharper resonance? (*Understand*) [2]

OR

- (ii) (a) An ac source of emf $e = 311 \sin(100\pi t)\ \text{V}$ is connected across a $110\ \Omega$ resistor. Find the rms voltage, the frequency and the average power dissipated. (*Apply*) [2]
- (b) State the principle of an ac generator. (*Recall*) [1]
- (c) An ideal transformer steps $2200\ \text{V}$ down to $220\ \text{V}$ and delivers $4.4\ \text{kW}$. Find its turns ratio and the currents in the primary and secondary. (*Apply*) [2]

Question 19

- (i) (a) Using Bohr's postulates, obtain an expression for the orbital speed of the electron in the n th orbit of a hydrogen atom. Show that its kinetic energy is numerically equal to its total energy. (*Apply*) [3]
- (b) Calculate the binding energy per nucleon of $^{20}_{10}\text{Ne}$. Atomic mass of $^{20}_{10}\text{Ne} = 19.992397\ \text{u}$. (*Apply*) [2]

OR

- (ii) (a) An α -particle of kinetic energy K is fired head-on at a nucleus of atomic number Z . Obtain an expression for its distance of closest approach and calculate it for $K = 5\ \text{MeV}$ and a gold nucleus ($Z = 79$). (*Apply*) [3]

- (b) Calculate the longest wavelength in the Lyman series of hydrogen. In which region of the spectrum does it lie? (*Apply*) [2]

Question 20

Read the passage and answer the questions that follow.

For a school astronomy night, Arjun builds a refracting telescope from two convex lenses: L_1 of power +2 D and aperture 8 cm, and L_2 of power +25 D and aperture 2 cm. His teacher explains that the great research telescopes use a concave mirror in place of the front lens.

- (i) Which lens should Arjun use as the objective? Give two reasons. (*Analyse*) [1]
- (ii) Draw a labelled ray diagram of the telescope in normal adjustment. (*Understand*) [2]
- (iii) Calculate the magnifying power and the length of the telescope in normal adjustment. (*Apply*) [1]
- (iv) Give one reason why large research telescopes use a concave mirror as the objective. (*Recall*) [1]

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